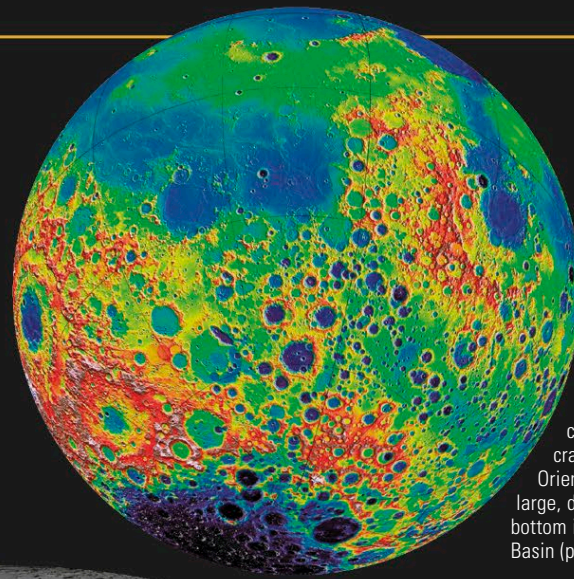
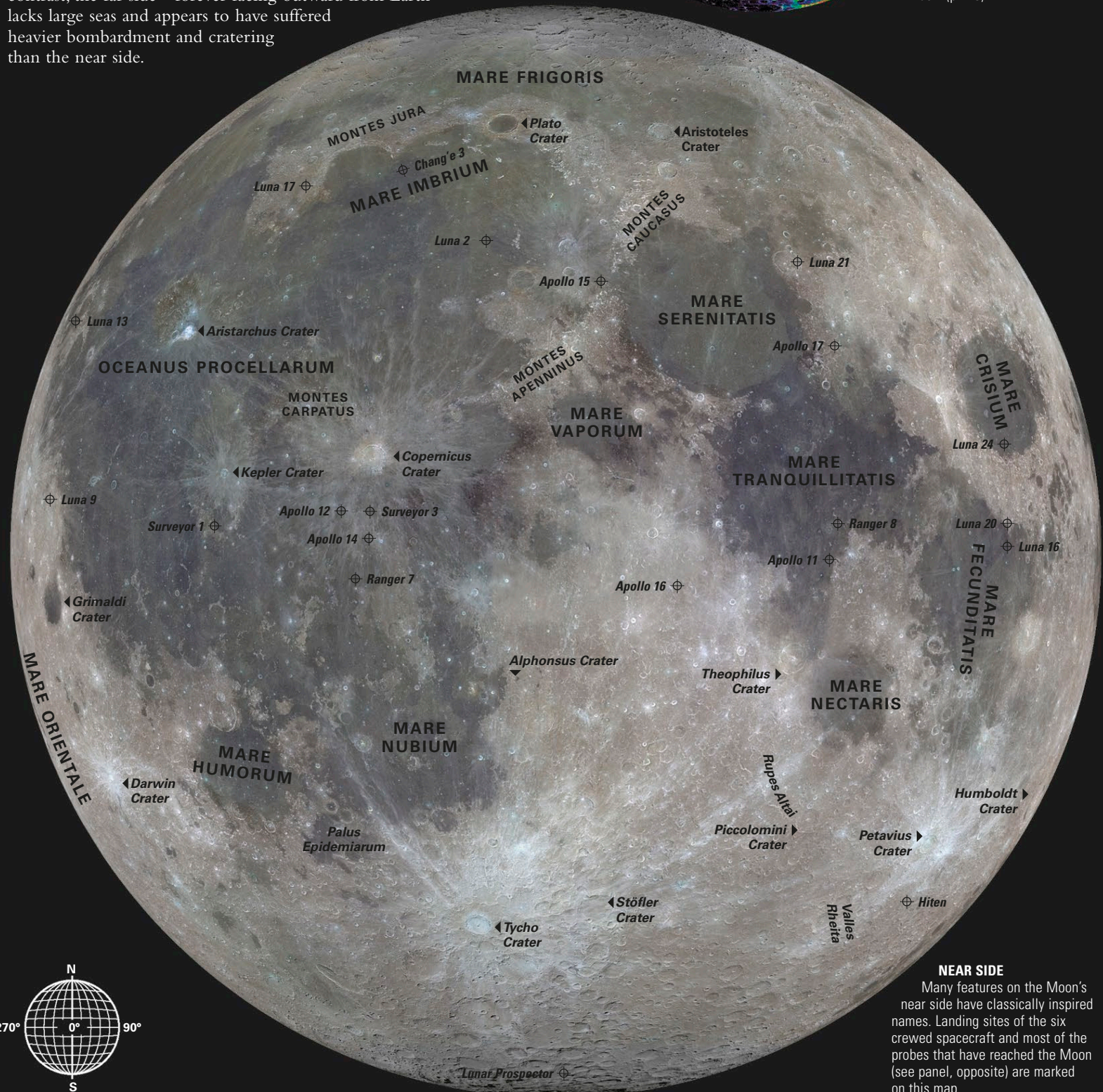


THE NEAR AND FAR SIDES OF THE MOON

The Moon's spin and orbital periods became locked together very early in its existence, when it was much closer to Earth than it is now and the surface was still molten due to the heating produced by massive early impacts. As a result, the Earth's influence has led to noticeable differences in the appearance of the two sides. The far side is on average about 3 miles (5 km) higher with respect to the Moon's center of mass than the near side, and its low-density crust is 16 miles (26 km) thicker. Because the near side is lower, volcanic magma has more easily found its way to the surface here, pouring from volcanic fissures into the low-lying regions of the largest craters and solidifying to form the lunar seas. By contrast, the far side—forever facing outward from Earth—lacks large seas and appears to have suffered heavier bombardment and cratering than the near side.



TOPOGRAPHIC VIEW
This map of lunar surface relief, color-coded according to height, was produced by the altimeter onboard NASA's Lunar Reconnaissance Orbiter. Red areas are the highest; blue areas the lowest. This view is centered on the bright ray crater Tycho, with the Mare Orientale basin at left. The large, dark blue feature at the bottom is the South Pole-Aitken Basin (p.149).



NEAR SIDE
Many features on the Moon's near side have classically inspired names. Landing sites of the six crewed spacecraft and most of the probes that have reached the Moon (see panel, opposite) are marked on this map.